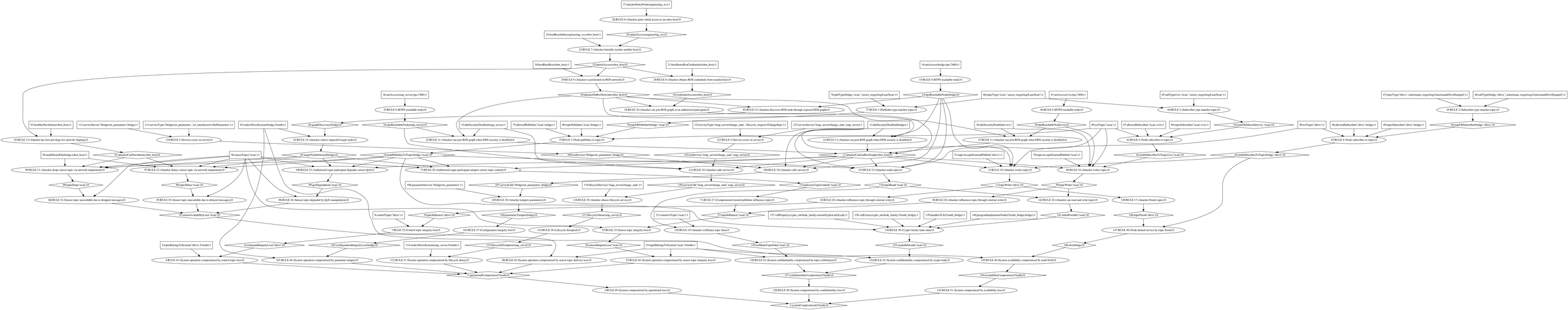

27:attackerEntryPoint(engineering_ws):1
26:RULE 6 (Attacker gains initial access to an entry host):0
24:hostReachable(engineering_ws,robot_host):1
25:initialAccess(engineering_ws):0
23:RULE 7 (Attacker laterally reaches another host):0
30:hostRunsRos(robot_host):1
22:lateralAccess(robot_host):0
21:hostStoresRosCredentials(robot_host):1
16:netAccess(bridge,rtps,7400):1
29:RULE 9 (Attacker is positioned on ROS network):0
20:RULE 8 (Attacker obtains ROS credentials from reached host):0
15:RULE 0 (RTPS reachable node):0
36:netAccess(map_server,rtps,7400):1
28:attackerOnRosNetwork(robot_host):0
19:credentialAccess(robot_host):0
78:pubType(bridge,’/scan’,’sensor_msgs/msg/LaserScan’):1
14:rtpsReachableNode(bridge):0
66:topicType(’/scan’,’sensor_msgs/msg/LaserScan’):1
41:netAccess(rviz,rtps,7400):1
65:subType(rviz,’/scan’,’sensor_msgs/msg/LaserScan’):1
47:topicType(’/drive’,’ackermann_msgs/msg/AckermannDriveStamped’):1
46:subType(bridge,’/drive’,’ackermann_msgs/msg/AckermannDriveStamped’):1
35:RULE 0 (RTPS reachable node):0
18:RULE 10 (Attacker can join ROS graph as an authorized participant):0
85:RULE 13 (Attacker discovers ROS node through exposed DDS graph):0
77:RULE 1 (Publisher type matches topic):0
40:RULE 0 (RTPS reachable node):0
64:RULE 2 (Subscriber type matches topic):0
45:RULE 2 (Subscriber type matches topic):0
93:hostHasNetAdmin(robot_host):1
112:serviceServer(’/bridge/set_parameters’,bridge):1
111:serviceType(’/bridge/set_parameters’,’rcl_interfaces/srv/SetParameters’):1
83:nodeAffectsSystem(bridge,f1tenth):1
84:graphDiscovery(bridge):0
34:rtpsReachableNode(map_server):0
33:ddsSecurityDisabled(map_server):1
79:allowedPublisher(’/scan’,bridge):1
80:topicPublisher(’/scan’,bridge):1
76:typeOkPublisher(bridge,’/scan’):0
124:serviceType(’/map_server/change_state’,’lifecycle_msgs/srv/ChangeState’):1
125:serviceServer(’/map_server/change_state’,map_server):1
13:ddsSecurityDisabled(bridge):1
38:ddsSecurityDisabled(rviz):1
39:rtpsReachableNode(rviz):0
69:rosTopic(’/scan’):1
67:allowedSubscriber(’/scan’,rviz):1
68:topicSubscriber(’/scan’,rviz):1
63:typeOkSubscriber(rviz,’/scan’):0
50:rosTopic(’/drive’):1
48:allowedSubscriber(’/drive’,bridge):1
49:topicSubscriber(’/drive’,bridge):1
44:typeOkSubscriber(bridge,’/drive’):0
92:RULE 12 (Attacker has host privilege for network shaping):0
110:RULE 5 (Service exists on server):0
82:RULE 14 (Attacker selects impactful target node):0
32:RULE 11 (Attacker can join ROS graph when DDS security is disabled):0
75:RULE 3 (Node publishes to topic):0
123:RULE 5 (Service exists on server):0
31:RULE 11 (Attacker can join ROS graph when DDS security is disabled):0
37:RULE 11 (Attacker can join ROS graph when DDS security is disabled):0
62:RULE 4 (Node subscribes to topic):0
43:RULE 4 (Node subscribes to topic):0
94:nodeHostedOn(bridge,robot_host):1
91:attackerCanNetAdmin(robot_host):0
56:sensorTopic(’/scan’):1
81:targetNodeSelected(bridge):0
74:nodePublishesToTopic(bridge,’/scan’):0
109:rosService(’/bridge/set_parameters’,bridge):0
122:rosService(’/map_server/change_state’,map_server):0
17:attackerCanJoinRosGraph(robot_host):0
51:topicAcceptsExternalPublish(’/drive’):1
70:topicAcceptsExternalPublish(’/scan’):1
61:nodeSubscribesToTopic(rviz,’/scan’):0
42:nodeSubscribesToTopic(bridge,’/drive’):0
90:RULE 21 (Attacker drops sensor topic via network impairment):0
97:RULE 22 (Attacker delays sensor topic via network impairment):0
100:RULE 23 (Authorized rogue participant degrades sensor QoS):0
73:RULE 20 (Authorized rogue participant tampers sensor topic content):0
121:RULE 18 (Attacker calls service):0
108:RULE 18 (Attacker calls service):0
133:RULE 15 (Attacker reads topic):0
12:RULE 16 (Attacker writes topic):0
60:RULE 16 (Attacker writes topic):0
89:topicDrop(’/scan’):0
96:topicDelay(’/scan’):0
99:qosDegradation(’/scan’):0
106:parameterService(’/bridge/set_parameters’):1
107:serviceCall(’/bridge/set_parameters’,bridge):0
119:lifecycleService(’/map_server/change_state’):1
120:serviceCall(’/map_server/change_state’,map_server):0
72:maliciousTopicContent(’/scan’):0
132:topicRead(’/scan’):0
11:topicWrite(’/drive’):0
59:topicWrite(’/scan’):0
88:RULE 34 (Sensor topic unavailable due to dropped messages):0
95:RULE 35 (Sensor topic unavailable due to delayed messages):0
98:RULE 36 (Sensor topic degraded by QoS manipulation):0
105:RULE 28 (Attacker tampers parameters):0
118:RULE 29 (Attacker abuses lifecycle service):0
71:RULE 27 (Compromised trusted publisher influences topic):0
10:RULE 26 (Attacker influences topic through external write):0
58:RULE 26 (Attacker influences topic through external write):0
142:RULE 24 (Attacker can read and write topic):0
149:RULE 17 (Attacker floods topic):0
87:sensorAvailabilityLoss(’/scan’):0
8:controlTopic(’/drive’):1
9:topicInfluence(’/drive’):0
104:parameterTamper(bridge):0
117:lifecycleAbuse(map_server):0
131:sensitiveTopic(’/scan’):1
57:topicInfluence(’/scan’):0
137:vulProperty(crypto_infoleak_family,remoteExploit,infoLeak):1
138:vulExists(crypto_infoleak_family,f1tenth_bridge):1
139:handlesTLS(f1tenth_bridge):1
140:programImplementsNode(f1tenth_bridge,bridge):1
141:mitmPossible(’/scan’):0
148:topicFlood(’/drive’):0
7:RULE 32 (Control topic integrity lost):0
103:RULE 37 (Configuration integrity lost):0
116:RULE 38 (Lifecycle disrupted):0
55:RULE 33 (Sensor topic integrity lost):0
130:RULE 19 (Attacker exfiltrates topic data):0
136:RULE 30 (Crypto family leaks data):0
147:RULE 40 (Node denied service by topic flood):0
5:topicBelongsToSystem(’/drive’,f1tenth):1
6:commandIntegrityLoss(’/drive’):0
102:configurationIntegrityLoss(bridge):0
114:nodeAffectsSystem(map_server,f1tenth):1
115:lifecycleDisruption(map_server):0
54:sensorIntegrityLoss(’/scan’):0
53:topicBelongsToSystem(’/scan’,f1tenth):1
129:exfiltrateTopicData(’/scan’):0
135:cryptoInfoLeak(’/scan’):0
146:dos(bridge):0
4:RULE 43 (System operation compromised by control topic loss):0
101:RULE 46 (System operation compromised by parameter tamper):0
113:RULE 47 (System operation compromised by lifecycle abuse):0
86:RULE 45 (System operation compromised by sensor topic delivery loss):0
52:RULE 44 (System operation compromised by sensor topic integrity loss):0
128:RULE 41 (System confidentiality compromised by topic exfiltration):0
134:RULE 42 (System confidentiality compromised by crypto leak):0
145:RULE 48 (System availability compromised by node DoS):0
3:operationalCompromise(f1tenth):0
127:confidentialityCompromise(f1tenth):0
144:availabilityCompromise(f1tenth):0
2:RULE 49 (System compromised by operational loss):0
126:RULE 50 (System compromised by confidentiality loss):0
143:RULE 51 (System compromised by availability loss):0
1:systemCompromised(f1tenth):0